Notes: Charoenwong, Kowaleski, Kwan & Sutherland (JFE, 2024)

RegTech: Technology-driven compliance and its effects on profitability, operations, and market structure

Contents

1 Big picture

- **Question.** What happens when financial regulation forces firms to invest in technology for compliance? Do these investments only raise compliance costs, or do they also change operations and market structure?
- **Setting.** The paper studies U.S. broker-dealers (BDs) around amendments to Rule 17a-5. The amendments require carrying BDs, which hold customer assets, to provide internal-control attestation over rules concerning capital and segregation of customer assets.
- **Treatment.** Carrying BDs are treated; noncarrying BDs are controls. The post period starts in 2014.
- **Core empirical design.** Difference-in-differences with BD fixed effects, FINRA district-by-year fixed effects, and lagged controls for size, business model, and advisor mix.
- **Main direct effect.** Treated BDs substantially increase RegTech investments: ERP adoption, servers, and technology-based compliance jobs rise after the amendment.
- **Profitability effect.** IT budgets rise and profitability falls, especially for smaller BDs and BDs with less sophisticated pre-period technology.
- **Complementarity effect.** Because information systems and data are nonrival inputs, compliance technology also enables noncompliance tools: communications management, CRM websites, and premium websites.
- **Operational effect.** Customer complaints and customer-reported misconduct decline after technology adoption. IV evidence using a Tech Index links the decline to technological investment.
- **Market-structure effect.** Acquisition activity and employee switching increase, and market concentration rises among carrying BDs.
- **Economic takeaway.** RegTech is not just a narrow compliance input. Regulation-induced technology adoption changes costs, service quality, employee mobility, acquisitions, and concentration in the financial sector.

2 Data and measurement

2.1 Broker-dealer setting and Rule 17a-5

- Broker-dealers are financial intermediaries regulated by the SEC and FINRA.
- Carrying BDs maintain custody of customer assets; noncarrying BDs do not.
- The 2013 amendment to Rule 17a-5 requires carrying BDs to maintain and document controls demonstrating moment-to-moment compliance with capital and customer-asset segregation requirements.
- The post period begins in 2014, after the amendment.

Interpretation: The amendment creates a clean compliance shock. It affects back-office control systems at carrying BDs but leaves noncarrying BDs as a natural comparison group.

2.2 Main datasets

- **Aberdeen Computer Intelligence Database:** firm-level software, hardware, and IT budget information.
- **BuiltWith:** website technology adoption, including CRM and other customer-facing technologies.
- **Revelio Labs:** job histories and job titles, used to measure RegTech, tech-based RegTech, and compliance jobs.
- **BrokerCheck:** customer complaints and customer-reported misconduct involving individual employees.
- **FINRA and BD filings:** business model, location, carrying status, and control variables.

Interpretation: The paper's contribution partly comes from assembling data on technology, jobs, complaints, operations, and market structure for both public and private financial institutions.

2.3 Technology investment variables

The direct RegTech outcomes include:

- **Adopts ERP:** indicator for first adoption of enterprise resource planning software.
- **Adds Additional ERP:** indicator for adding another ERP program when the BD already used ERP.
- **Servers:** number of servers in Aberdeen hardware data.
- **RegTech Jobs:** share of jobs with RegTech-related titles.
- **Tech-based RegTech Jobs:** share of jobs that are both technology and RegTech related.
- **Compliance Jobs:** share of jobs in traditional compliance roles.

Interpretation: ERP and servers proxy for large fixed technology investments. RegTech jobs capture human capital complementary to the technology investment.

2.4 Profitability and IT budget

- **IT Budget** is measured as 100 times the log of IT budget.
- **Profitability** is net income divided by average pre-amendment capital, multiplied by 100.
- Effects are studied both for the full sample and by BD size/technology sophistication.

Interpretation: If compliance technology is costly, IT budgets should increase and profitability should decline. If the cost has fixed components, smaller or less technologically prepared BDs should be hit harder.

2.5 Complementary technology variables

The paper studies customer-facing and operational technology that can be built on top of improved information systems:

- communications management software;
- CRM website technologies;
- premium website technologies;
- job application and payroll software as a placebo/noncustomer technology.

Interpretation: The central complementarity claim is that RegTech investments improve the firm's information environment, making it easier to deploy other digital tools.

2.6 Complaints and misconduct

- **Complaint** equals 100 times an indicator for a customer complaint in BrokerCheck.
- **Customer-reported misconduct** equals 100 times an indicator for a customer-reported misconduct incident.
- **Complaint** $>$ \$5000 measures more costly complaints.

Interpretation: These outcomes are used to test whether technology improves monitoring and customer service. The paper distinguishes actual service improvements from regulator/auditor attention effects.

2.7 Market-structure variables

- **Acquisition** equals 100 times an indicator that a BD conducts an acquisition.
- **Has Switcher** equals 100 times an indicator that a BD hires an employee from another specific BD.
- **HHI among carrying BDs** measures county-year concentration among carrying BDs.
- **Number of carrying BDs** counts carrying BDs in a county.

Interpretation: These outcomes measure whether compliance costs and complementary technology investments change industrial organization in the broker-dealer market.

3 Models and empirical specifications

3.1 Baseline difference-in-differences specification

The main regression is:

$$y_{i,t} = \alpha_i + \alpha_{f(i,t)} + \beta \, \text{Post}_t \times \text{Treated}_i + \Gamma' X_{i,t-1} + \varepsilon_{i,t}. \tag{1}$$

- i indexes BDs and t indexes years.
- Post_t equals one starting in 2014.
- Treated_i equals one for carrying BDs.
- α_i are BD fixed effects.
- $\alpha_{f(i,t)}$ are FINRA district-by-year fixed effects.
- $X_{i,t-1}$ includes lagged size, age, tenure, dual-registration share, and complaint history.

Interpretation: The coefficient β compares the post-2014 change at carrying BDs to the contemporaneous change at noncarrying BDs in the same FINRA district-year.

3.2 Event-time version

The paper also estimates event-time specifications of the form:

$$y_{i,t} = \alpha_i + \alpha_{f(i,t)} + \sum_{k \neq 2010} \beta_k \mathbf{1}\{t = k\} \times \text{Treated}_i + \Gamma' X_{i,t-1} + \varepsilon_{i,t}. \tag{1}$$

Interpretation: Event-time plots show whether treated and control BDs had similar pre-trends and whether effects appear around the amendment.

3.3 Tech Index and IV logic

The paper constructs a **Tech Index** using the average standardized values of ERP investments, servers, and CRM website technologies. It then uses treatment as a source of variation in technology:

$$TechIndex_{i,t} = \pi_0 + \pi_1 \text{Post}_t \times \text{Treated}_i + \text{FE} + X'_{i,t-1}\Pi + u_{i,t}, \tag{2}$$

$$Complaint_{i,t} = \rho_0 + \rho_1 \widehat{TechIndex}_{i,t} + \text{FE} + X'_{i,t-1}P + e_{i,t}. \tag{3}$$

Interpretation: This IV exercise asks whether the complaint decline operates through technology investment rather than through direct regulator or auditor attention.

3.4 Employee switching specification

For pairs of origin and destination BDs, the employee-switching regression can be summarized as:

$$Switch_{i,j,t} = \alpha_{i,j} + \mu_t + \beta \, Type_{i,j} \times \text{Post}_t + \varepsilon_{i,j,t}. \tag{2}$$

- $Switch_{i,j,t}$ indicates whether an employee moves from BD i to BD j.
- Pair fixed effects absorb persistent relationships between BDs.
- The key variables distinguish moves from noncarrying to carrying BDs, carrying to noncarrying BDs, and carrying to carrying BDs.

Interpretation: This specification tests whether technological complementarities and acquisition activity reallocate labor toward treated/carrying BDs.

3.5 Market concentration specification

The concentration tests use county-year outcomes:

$$HHI_{c,t} = \alpha_c + \mu_t + \beta\, HighExposure_c \times \mathrm{Post}_t + \eta_{c,t}. \tag{4}$$

Interpretation: The county exposure is based on the pre-amendment share of local employment at carrying BDs. A positive β means concentration rises in places more exposed to the Rule 17a-5 compliance shock.

4 Identification and empirical design

4.1 What identifies the effects

- The amendment applies to carrying BDs but not to noncarrying BDs.
- The timing is sharp: the post period begins in 2014.
- BD fixed effects absorb time-invariant differences between carrying and noncarrying BDs.
- FINRA district-by-year fixed effects absorb local regulatory and economic conditions.
- Lagged controls absorb changes in size, age, advisor mix, and complaint history.

Interpretation: The identifying assumption is that, conditional on fixed effects and controls, carrying and noncarrying BDs would have followed similar trends absent the amendment.

4.2 Why technology mechanisms are credible

- The direct technology effects appear in ERP, servers, and tech-based RegTech jobs, which are plausibly needed for internal control documentation.
- Complementary technology effects occur in CRM and communication tools but not in the job application/payroll placebo.
- Complaint reductions are linked to a technology index in IV specifications.
- COVID-19 remote-work tests provide additional evidence that stronger pre-existing technology improves customer-service resilience.

Interpretation: The paper uses multiple pieces of evidence to connect regulation-induced technology investments to operational improvements.

4.3 Ruling out attention explanations

- If regulator attention drives complaint declines, declines should be stronger for retail-focused BDs and regulator-reported complaints.
- If auditor attention drives complaint declines, declines should be stronger where auditors are distant or less tolerant, and for auditor-related complaints.
- The paper generally finds patterns inconsistent with pure regulator or auditor attention explanations.

Interpretation: The complaint decline is more consistent with improved monitoring and customer service than with more external scrutiny.

4.4 What remains imperfect

- Carrying and noncarrying BDs differ in business models.
- Data coverage differs across datasets and favors larger BDs.
- Technology adoption is observed through databases that may miss some internal systems.
- The paper measures complaints and misconduct but not every dimension of service quality.
- Market concentration effects are plausibly related to fixed compliance costs, but acquisition decisions may also reflect other industry trends.

Interpretation: The paper's strength is the coherent set of outcomes around a clear regulatory shock, not a randomized technology adoption experiment.

5 Tables and figures

Visuals are directly extracted from the paper and grouped to save space.

5.1 Table 1: Summary statistics

Read:

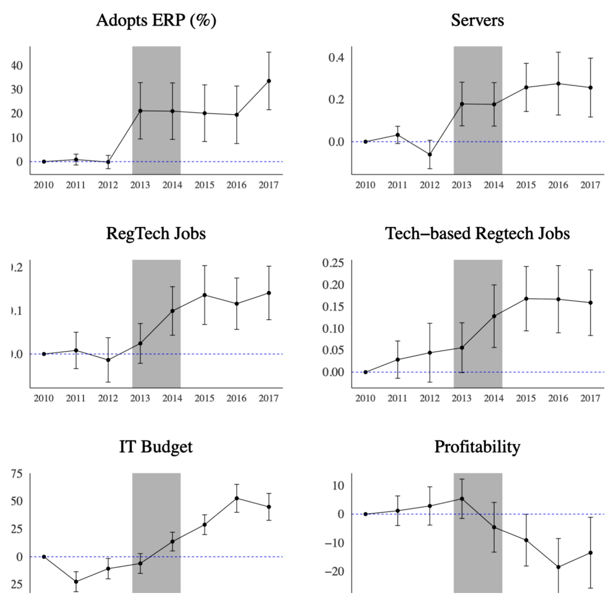
- Panel A summarizes BD characteristics. Treated BDs make up about 5.4% of observations, while post-amendment observations are 47.4% of the sample.
- Average BD size is highly skewed: mean total assets are much larger than the median.
- Panel B shows that preexisting RegTech investment is limited. Only 12.7% adopt ERP for the first time in the sample; server counts and IT budgets are also skewed.
- Revelio job measures show that RegTech and compliance-related jobs are small but measurable shares of employment.

Economic message: The sample captures a heterogeneous BD industry, where fixed compliance technology costs should matter most for smaller and less technologically sophisticated firms.

Table 1
Summary statistics.

Variable	Mean	SD	P25	Median	P75
Panel A: BD Characteristics					
BD Characteristics:					
Total Assets ($000 s)	1261,841	16,879,300	143	668	4883
Total Net Capital ($000 s)	647,952	87,408,383	61	293	1905
Treated	0.054	0.227	0	0	0
Post	0.474	0.499	0	0	1
Lag Num. Employees	211	1709	5	11	37
Lag Avg. Tenure (years)	6.219	5.311	2.600	4.800	8.027
Lag Fraction of Dual-Registered Employees	0.294	0.309	0.000	0.200	0.523
Fraction of Employees with Complaint History	0.045	0.101	0.000	0.000	0.041
Complaint Measures:					
1(Complaints > 0)	0.099	0.299	0	0	0
1(Customer-reported Misconduct> 0)	0.075	0.264	0	0	0
Financial Measures:					
Profitability	41.396	217.685	−13.557	5.454	38.39
Panel B: RegTech Investments					
Aberdeen Software:					
Adopts ERP (%)	12.689	33.292	0	0	0
Adds Additional ERP (%)	87.900	32.623	100	100	100
Aberdeen Hardware:					
Servers	470	2723	2	6	45
IT Budget ($000 s)	49,921	311,872	110	447	3990
Revelio:					
RegTech Jobs	0.098	0.130	0.000	0.062	0.136
Tech-based RegTech Jobs	0.051	0.096	0.000	0.012	0.061
Compliance Jobs	0.047	0.084	0.000	0.025	0.062

This table presents summary statistics for BD characteristics in Panel A and RegTech investment variables in Panel B. All observations are at the BD-year level. The BD characteristics sample has 26,721 BD-year observations from 4660 unique BDs. The profitability sample has 19,845 BD-year observations from 3990 unique BDs. The Aberdeen Software sample has 7694 BD-year observations from 2524 unique BDs. The Aberdeen Hardware sample has 18,313 BD-year observations from 3227 unique



5.2 Figure 1.1: RegTech investment, labor demand, and profitability

Read:

- ERP adoption rises sharply for carrying BDs around the amendment.
- Server use and RegTech-related jobs also increase.
- IT budgets rise after the amendment.
- Profitability falls after the amendment, especially after technology spending ramps up.

Economic message: The amendment induces real technology investment, not merely paperwork. But the investment is costly and lowers profitability.

5.3 Table 2 + Table 3: Direct technology adoption and profitability

Read: Table 2:

- Treated BDs are 16.3 percentage points more likely to adopt ERP for the first time after the amendment.
- Servers increase by 0.189 and RegTech jobs rise by 0.118.
- Tech-based RegTech jobs rise by 0.122, while traditional compliance jobs do not significantly increase.

Read: Table 3:

- IT budgets rise by 23.8 in the full sample and by 28.7 among small BDs.
- Profitability falls by about 13.7 percentage points in the full sample.
- The profitability decline is larger for smaller BDs and low-technology BDs.

Economic message: Regulation directly pushes firms toward RegTech adoption. The cost burden is large and concentrated among firms less able to absorb fixed technology costs.

Table 2
RegTech investments and labor demand.

Dep Var:	Adopts ERP (%) (1)	Adds Additional ERP (%) (2)	Servers (3)	RegTech Jobs (4)	Tech-based RegTech Jobs (5)	Compliance Jobs (6)
Treated × Post	16.332**	1.854	0.189**	0.118***	0.122**	0.074
	(7.583)	(1.429)	(0.081)	(0.039)	(0.048)	(0.055)
N	2921	3443	16,378	13,766	13,766	13,766
R^2	0.845	0.885	0.883	0.207	0.271	0.200
Treated Share	0.072	0.184	0.074	0.071	0.071	0.071
Mean of Dep Var	10.476	84.577	364.173	0.098	0.051	0.047
SD of Dep Var	30.629	36.122	1693.583	0.130	0.096	0.084
Sample	No ERP in pre-period	Has ERP in pre-period	Full	Full	Full	Full

This table studies RegTech investments and labor demand using Eq. (1). In column 1 (2), *Adopts ERP% (Adds Additional ERP%)* is 100 times an indicator variable for adopting ERP for the first time (adding an additional ERP program). In columns 3–6, the dependent variable is the number of servers or the share of each job type as labeled. *Post* is an indicator variable equal to one starting in 2014. *Treated* is an indicator variable equal to one for carrying BDs (static throughout the sample). Columns 1–2 (3, 4–6) use OLS (Poisson, fractional regression) estimation. The sample in column 1 (2) is limited to BDs without (with) ERP in the pre-amendment period.

Table 3
Profitability.

Dep Var:	IT Budget (1)	Profitability (2)	IT Budget (3)	IT Budget (4)	Profitability (5)	Profitability (6)	Profitability (7)	Profitability (8)
Treated × Post	23.843***	−13.659**	4.286	28.737***	0.246	−20.828*	−13.392*	−28.851***
	(6.168)	(6.586)	(8.086)	(9.949)	(4.368)	(11.551)	(7.532)	(11.007)
N	13,214	19,793	2259	10,676	2356	17,437	5804	5810
R^2	0.938	0.624	0.942	0.917	0.763	0.620	0.687	0.655
Treated Share	0.074	0.055	0.231	0.043	0.229	0.031	0.129	0.031
Mean of Dep Var	1361.518	40.327	1621.668	1307.702	13.653	43.584	26.865	49.197
SD of Dep Var	250.942	209.742	228.453	218.540	54.256	221.343	158.237	209.785
Sample	Full	Full	Large	Small	Large	Small	High IT	Low IT

This table studies profitability using Eq. (1). In columns 1, 3 and 4, the dependent variable is *IT Budget*, 100 times the log of the IT budget. In columns 2 and 5–8 the dependent variable is *Profitability*, the ratio of net income to average pre-amendment capital multiplied by 100. *Post* is an indicator variable equal to one starting in

5.4 Table 4 + Table 5: Complementary investments and customer outcomes

Read: Table 4:

- Carrying BDs become more likely to adopt communications management software after the amendment.
- Website CRM and premium website technologies rise by about 10.5 percentage points.
- The job application and payroll placebo is insignificant.

Read: Table 5:

- Customer complaints decline by 4.42 percentage points.
- Customer-reported misconduct declines by 3.58 percentage points.
- Complaints with damages above $5000 decline by 4.32 percentage points.
- IV estimates using the Tech Index point to technology as the channel.

Economic message: Compliance technology creates a platform for customer-facing technologies. These complementary tools improve monitoring and reduce complaints and misconduct.

Table 4
Complementary investments.

Dep Var:	Comm. Management (1)	Website CRM (2)	Website Premium (3)	Job App. And Payroll (4)
Treated × Post	3.005**	10.507***	10.573***	0.469
	(1.416)	(2.872)	(2.876)	(1.720)
N	6262	10,654	10,654	6262
R^2	0.965	0.290	0.314	0.832
Treated Share	0.133	0.094	0.094	0.133
Mean of Dep Var	25.515	18.315	20.412	24.501
SD of Dep Var	43.598	38.681	40.307	43.013

This table studies complementary investments using Eq. (1). The dependent variables in columns 1 and 4 are 100 times indicator variables for having the software type labeled in the column header. The dependent variables in columns 2 and 3 are 100 times indicators for adding a new website technology of the type labeled in the column header. *Post* is an indicator variable equal to one starting in 2014, and *Treated* is an indicator variable equal to one for carrying BDs.

Table 4: Complementary investments.

Table 5
Technological investment, customer complaints, and misconduct.

Panel A: Reduced-Form

Dep Var:	Complaint (1)	Customer-reported Misconduct (2)	Complaint >$5000 (3)
Treated × Post	−4.420***	−3.580**	−4.321***
	(1.482)	(1.391)	(1.536)
N	17,810	17,810	17,810
R^2	0.700	0.668	0.695
Treated Share	0.073	0.073	0.073
Mean of Dep Var	12.847	9.888	11.786
SD of Dep Var	33.462	29.981	32.245

Panel B: Instrumental Variables

Dep Var:	Tech Index (1)	Complaint (2)	Customer-reported Misconduct (3)	Complaint >$5000 (4)
Treated × Post	0.173***			
	(0.029)			
Tech Index		−25.557***	−20.704**	−24.984**
		(9.496)	(8.474)	(9.823)
F-Statistic	35.7			
N	17,810	17,810	17,810	17,810
R^2	0.866	0.654	0.634	0.649
Treated Share	0.073	0.073	0.073	0.073
Mean of Dep Var	0.000	12.847	9.888	11.786
SD of Dep Var	0.784	33.462	29.981	32.245

This table studies customer complaints and misconduct using Eq. (1). The dependent variable in Panel A, column 1 is 100 times an indicator variable for whether the BD has a customer complaint recorded on BrokerCheck that year. The dependent variable in column 2 is 100 times an indicator variable for whether the BD has a customer-reported misconduct incident that year. The dependent variable in column 3 is 100 times an indicator variable for whether the BD has a complaint with alleged damages of at least $5000. In Panel B, *Tech*

Table 5: Complaints and misconduct.

5.5 Figure 1.2 + Table 6: Complaints, misconduct, and attention tests

Read:

- Figure 1.2 shows CRM and communications management technologies rising after the amendment.
- Complaints and customer-reported misconduct decline after the amendment.
- Table 6 investigates regulator and auditor attention explanations.
- The complaint decline does not line up with the patterns predicted by simple regulator or auditor attention stories.

Economic message: The complaint decline appears to reflect improved technology-enabled monitoring rather than only increased external scrutiny.

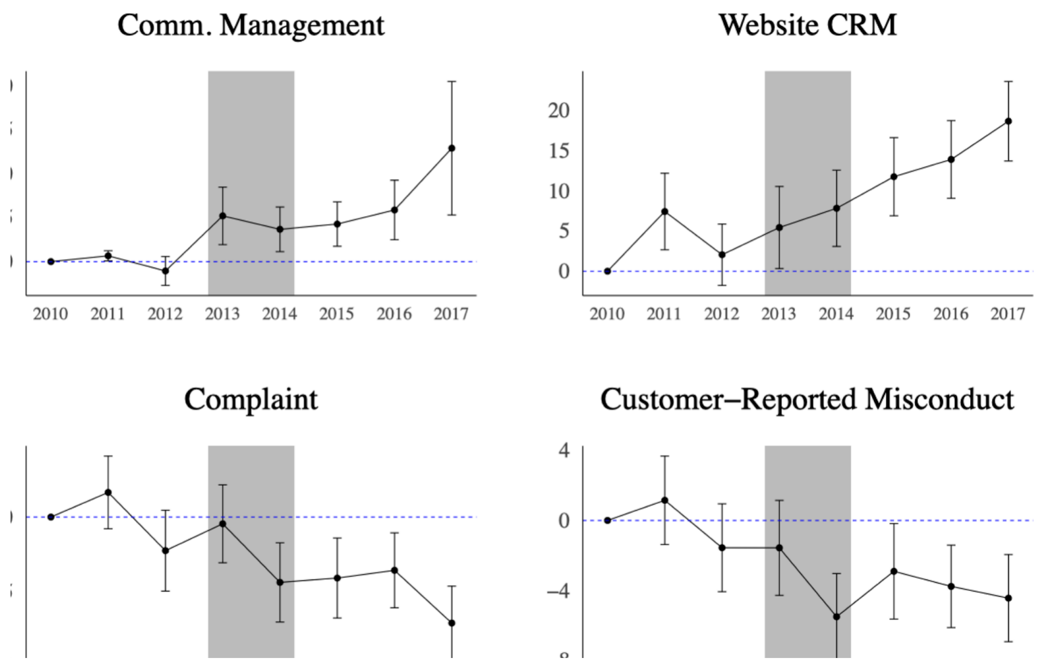


Figure 1.2: Complementary investments, complaints, and misconduct.

Table 6
Investigating regulator and auditor attention.

Dep Var:	Complaint (1)	Regulator-reported (2)	Regulator-reported (3)	Complaint (4)	Complaint (5)	Auditor-related Complaint (6)	Not Auditor-related Complaint (7)
Treated × Post	−0.895	0.931	−0.468	−4.108**	−5.286***	0.056	−4.710***
	(1.376)	(1.933)	(2.395)	(1.948)	(2.017)	(1.303)	(1.460)
Treated × Post × Retail	−4.986**						
	(2.434)						
Treated × Post × Distant			2.683		2.170		
			(3.710)		(2.868)		
Treated × Post × Auditor Tolerant				−0.323			
				(3.024)			
N	17,810	10,944	10,944	16,035	17,802	17,810	17,810
R^2	0.699	0.492	0.492	0.703	0.699	0.503	0.702
Treated Share	0.073	0.083	0.083	0.077	0.073	0.073	0.073
Mean of Dep Var	12.847	10.124	10.124	13.096	12.847	3.330	12.487
SD of Dep Var	33.462	30.166	30.166	34.737	33.462	17.941	33.058
Sample	Full	Retail	Retail	Full	Full	Full	Full

This table studies complaints using Eq. (1). The dependent variable is 100 times an indicator variable for whether the BD has a complaint of the type labeled in the column header recorded on BrokerCheck that year. *Post* is an indicator variable equal to one starting in 2014, and *Treated* is an indicator variable equal to one for carrying BDs. *Retail* is an indicator variable for whether the BD offers retail-facing products, including investment advice, mutual funds, variable life insurance, and debt products. In column 3 (5), *Distant* is an indicator variable for whether the BD is farther than the median from its nearest FINRA office (from its auditor's office). *Auditor Tolerant* is an indicator for the BD's auditor having a size-adjusted above-median share of clients with a customer complaint recorded on BrokerCheck between 2010 and 2013. The sample in columns 2 and 3 is limited to Retail BDs. All regressions include controls from Eq. (1) and BD and FINRA district-by-year fixed effects. Standard errors are clustered by BD and shown in parentheses. * signifies $p < 0.1$, ** signifies $p < 0.05$, and *** signifies $p < 0.01$. At the bottom of the table, we report

Table 6: Regulator and auditor attention.

5.6 Figure 1.3 + Tables 7–8: Consolidation, switching, and concentration

Read: Figure 1.3:

- Acquisition activity rises for large carrying BDs with high pre-amendment IT.
- Acquisition activity does not rise meaningfully for small BDs.
- Employee switching into carrying BDs rises after the amendment.
- HHI among carrying BDs increases, but HHI among noncarrying BDs does not.

Read: Tables 7–8:

- Table 7 shows a higher probability that employees leave noncarrying BDs to join carrying BDs after the amendment.
- Table 8 shows nontrivial concentration increases among carrying BDs in high-exposure counties.
- The number of carrying BDs declines in high-exposure counties.

Economic message: Fixed RegTech costs and technology complementarities reshape market structure: larger technologically prepared firms expand, workers move, and concentration rises.

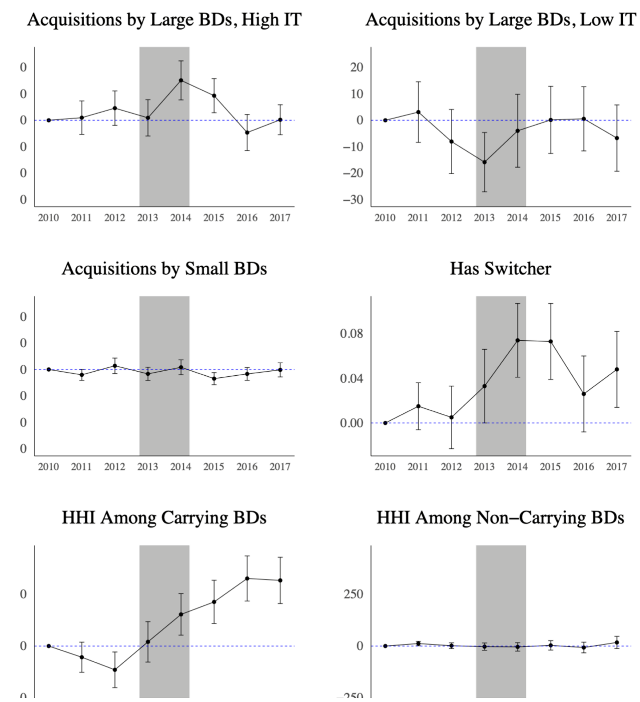


Figure 1.3: Consolidation, switching, and market concentration.

Table 7
Employee switching.

Dep Var:	Has Switcher (1)	Has Switcher (2)	Has Switcher (3)
Leaves Noncarrying, Joins Carrying × Post	0.038**	0.041**	0.041**
	(0.016)	(0.016)	(0.016)
Leaves Carrying, Joins Noncarrying × Post		0.024	
		(0.015)	
Leaves Carrying x Joins Carrying × Post		0.096	
		(0.096)	
N	53,595,473	53,595,473	53,595,473
R^2	0.387	0.387	0.388
Mean Dep Var	0.163	0.163	0.163
SD Dep Var	4.029	4.029	4.029

This table studies employee switching using Eq. (2). The dependent variable is 100 times an indicator variable for whether the BD has an employee join from another specific BD that year, e.g., BD_i from BD_j. The independent variables are indicators for combinations of types of origin and destination BDs for the employee, times *Post*, an indicator variable equal to one starting in 2014. The sample includes all pairs of destination and origin BDs involving BDs with at least one employee switch during the sample window. Observations are at the BD firm pair-year level. Columns 1 and 2 include origin-by-destination BD-pair and year fixed effects, and column 3 adds origin-by-year fixed effects. Standard errors are clustered by BD_i and BD_j and shown in parentheses. * signifies $p < 0.1$,

Table 7: Employee switching.

Table 8
Market concentration.

Panel A: HHI among Carrying BDs			
Dep Var:	HHI		
High Exposure Share Threshold =	Median	75th Percentile	90th Percentile
	(1)	(2)	(3)
High Exposure × Post	292.527***	240.494***	268.672***
	(65.818)	(61.475)	(87.479)
N	8800	8800	8800
R^2	0.855	0.854	0.854
Mean of Dep Var	3468.320	3468.320	3468.320
SD of Dep Var	2087.640	2087.640	2087.640
Num. Treated Counties	671	228	53
Panel B: HHI among Noncarrying BDs			
Dep Var:	HHI		
High Exposure Share Threshold =	Median	75th Percentile	90th Percentile
	(1)	(2)	(3)
High Exposure × Post	−11.420	20.296	41.841
	(18.047)	(23.507)	(47.201)
N	8800	8800	8800
R^2	0.938	0.938	0.938
Mean of Dep Var	1244.835	1244.835	1244.835
SD of Dep Var	838.210	838.210	838.210
Panel C: Number of Carrying BDs			
Dep Var:	Number of Carrying BDs		
High Exposure Share Threshold =	Median	75th Percentile	90th Percentile
	(1)	(2)	(3)
High Exposure × Post	−0.070***	−0.059***	−0.091***
	(0.008)	(0.010)	(0.021)
N	8776	8776	8776
R^2	0.992	0.992	0.992
Mean of Dep Var	8.230	8.230	8.230
SD of Dep Var	9.103	9.103	9.103

This table studies market concentration. The dependent variable in Panels A and B is the Herfindahl-Hirschman index for the county-year, where the index is based on headcount and spans [0,10,000]. The dependent variable in Panel C is

Table 8: Market concentration.

6 Short summary

- The paper studies Rule 17a-5 amendments that impose internal-control attestation requirements on carrying broker-dealers.
- Treated BDs invest in ERP, servers, and tech-based RegTech jobs.
- IT budgets rise and profitability declines, especially at smaller and less technologically sophisticated BDs.
- Compliance technology enables complementary investments in communications management, CRM, and premium websites.
- Customer complaints and customer-reported misconduct decline.
- IV evidence using a Tech Index points to technology as the mechanism behind the complaint decline.
- Regulator/auditor attention explanations do not fully fit the pattern of results.
- Acquisition activity rises among large, high-IT carrying BDs.
- Employees move from noncarrying to carrying BDs.
- Market concentration rises among carrying BDs in high-exposure counties.
- The broad lesson is that RegTech changes costs, operations, and industrial organization, not just compliance documentation.

7 Conclusion

7.1 Core conclusion

Charoenwong, Kowaleski, Kwan, and Sutherland show that compliance-driven technology investments have broad consequences for the financial sector. New internal-control requirements lead carrying BDs to adopt ERP, expand servers, and hire technology-based RegTech workers. These investments raise IT budgets and reduce profitability, especially for smaller firms.

7.2 Economic intuition

Regulation forces firms to upgrade internal information systems. Those systems are costly, but they also create a platform for complementary tools such as CRM and communications management software. These tools can improve monitoring and customer service, reducing complaints and misconduct. At the same time, fixed costs and complementary investment needs favor larger, technologically sophisticated firms.

7.3 Takeaway

RegTech is both a compliance response and a source of organizational change. The welfare consequences of financial regulation therefore depend not only on direct compliance costs, but also on how technology reshapes operations, labor mobility, acquisitions, and market concentration.